

Adaptive Energy Management Strategy for Battery–Fuel Cell Hybrid Electric Vehicle with XGBoost-Based Excess and Deficit Power Mode Classification

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Abstract— Efficient energy management is a significant requirement in fuel cell-battery Hybrid Electric Vehicles (HEVs), which is essential for efficient power distribution, high efficiency, and component life. The conventional energy management strategies, such as rule-based and fuzzy logic control, are effective in managing the energy in HEVs, but these control strategies may not be adaptive to changing conditions. This paper proposes an adaptive energy management strategy for battery-fuel cell-based HEVs using a fuzzy logic-based energy management strategy and machine learning. Initially, a fuzzy logic controller is designed using MATLAB, which manages the energy distribution between the battery and fuel cell in accordance with the varying demand conditions. The operational data from the fuzzy logic-based energy management strategy is used to develop an extreme gradient boosting model, which predicts the instantaneous power demand and classifies the operation into two modes: Excess Power and Deficit Power. The proposed strategy improves the intelligence of the energy management system, which can efficiently manage the energy distribution in accordance with the instantaneous demand conditions, thus improving the overall efficiency of the battery and fuel cell in HEVs.

Keywords— Hybrid Electric Vehicles (HEVs), Energy Management Strategy (EMS), Fuel Cell–Battery Hybrid System, Fuzzy Logic Controller, Extreme Gradient Boosting (XGBoost), Power Demand Prediction.

I. INTRODUCTION

Hybrid Electric Vehicles (HEVs) have emerged as a promising technology to address the increasing demand for energy efficiency and the reduction of greenhouse gas emissions in the transportation sector. The combination of multiple energy sources, such as batteries and fuel cells, allows HEVs to achieve improved efficiency, reliability, and environmental performance compared with conventional internal combustion engine vehicles. Fuel cells provide clean and continuous energy generation with high efficiency, while batteries offer high power density

and rapid response to transient power demands. However, the effective coordination of these energy sources requires an efficient energy management system (EMS) capable of distributing power according to the vehicle's operating conditions [1], [2].

Energy management strategies play a vital role in optimizing the performance of hybrid electric vehicles by controlling the power flow between the fuel cell, battery, and vehicle load. Several approaches have been developed to address this challenge, including rule-based control, optimization-based methods, and intelligent control techniques. Among these, rule-based strategies are widely used because of their simplicity and ease of implementation. However, these strategies often rely on predefined rules and may not perform optimally under varying driving conditions or dynamic power demands [3]. To overcome these limitations, intelligent control methods such as fuzzy logic control have been introduced. Fuzzy logic controllers are capable of handling nonlinear system behavior and uncertainties, making them suitable for hybrid energy systems [4], [5].

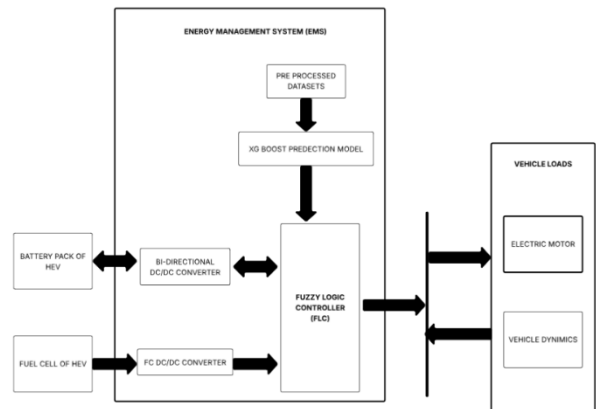


Fig. 1. A block diagram illustrating the hybrid electric vehicle energy management system showing the fuel cell, battery, fuzzy logic controller, XGBoost prediction model, and power flow to the vehicle load.

Despite their advantages, conventional fuzzy logic-based control strategies still face challenges in adapting to rapidly changing driving conditions and unpredictable load variations. In recent years, machine learning techniques have gained significant attention in energy management applications due to their ability to learn patterns from operational data and improve decision-making processes [6]. Advanced algorithms such as Extreme Gradient Boosting (XGBoost) have demonstrated strong performance in prediction and classification tasks, particularly in complex energy systems and smart grid applications [7], [8].

By integrating machine learning with traditional control methods, more adaptive and intelligent energy management strategies can be developed. Recent studies have shown that data-driven models can enhance the performance of hybrid energy systems by predicting power demand and optimizing energy distribution between different sources [9], [10]. Therefore, combining fuzzy logic control with machine learning techniques offers a promising approach to improving the efficiency and adaptability of energy management systems in fuel cell-battery hybrid electric vehicles as shown in Fig 1.

II. RELATED WORKS

A. Battery System Design in MATLAB

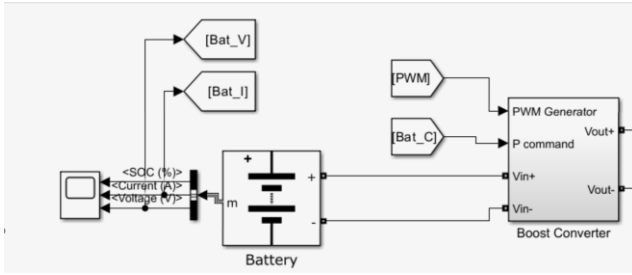


Fig. 2. Developed simulink model of Battery with DC-DC(Boost) converter controlled by Fuzzy logic controller

The battery system plays a critical role in hybrid electric vehicles by providing immediate power support during transient conditions and storing excess energy for later use. In this work, the battery model is developed in the MATLAB/Simulink environment using a controlled voltage source combined with internal resistance and state-of-charge (SOC) dynamics. The model is designed to accurately reflect the real-time behavior of lithium-ion batteries, including charge - discharge characteristics and voltage variation with respect to SOC.

$$V_{bat} = V_{oc} (SOC) - I_{bat} \cdot R_{int}$$

The battery is interfaced with the DC bus of the vehicle through a bidirectional DC-DC converter, which enables both charging and discharging operations. During high power demand conditions, the converter operates in boost mode to supply additional energy from the battery to the DC bus. Conversely, during low demand or excess power conditions, it operates in buck mode to store surplus energy back into the battery. This bidirectional energy flow ensures efficient utilization of stored energy and improves overall system performance.

The battery model is further integrated with the energy management system, where the control signals generated by the fuzzy logic controller determine the power contribution of the battery. This approach allows the battery to respond

dynamically to varying load conditions, thereby enhancing system stability and extending battery lifespan.

The state of charge is dynamically updated using:

$$SOC = SOC_{initial} - \frac{1}{Q} \int I_{bat} dt$$

To enable efficient energy exchange, the battery is connected to the DC bus through a bidirectional DC-DC converter. The converter operates in boost mode when supplying power to the load and in buck mode during charging.

The power delivered by the battery is given by:

$$P_{bat} = V_{bat} \cdot I_{bat}$$

This configuration allows the battery to respond quickly to load variations, thereby supporting peak demand conditions. Similar battery modeling and control approaches have been discussed in the literature [4], [9].

B. Fuel Cell System Design in MATLAB

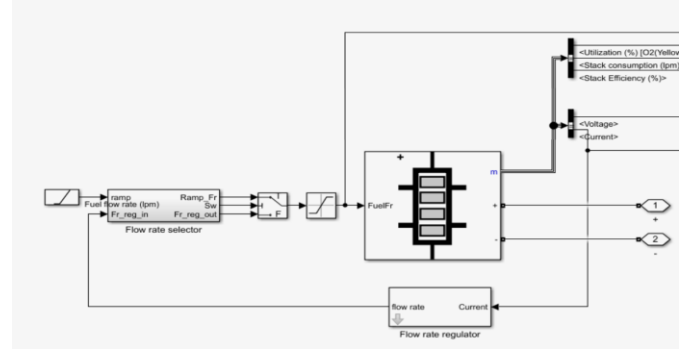


Fig. 3. Developed simulink model of Fuel cell DC-DC(Buck-Boost) converter controlled by Fuzzy logic controller

The fuel cell serves as the primary energy source in the hybrid system due to its high efficiency and capability of delivering sustained power. In this work, a proton exchange membrane (PEM) fuel cell model is implemented in MATLAB/Simulink. The model captures the essential characteristics of the fuel cell, including voltage-current behavior, efficiency, and response to load variations.

Since the output voltage of the fuel cell is typically low and varies with operating conditions, a unidirectional DC-DC boost converter is employed to step up the voltage to the required DC bus level. This ensures compatibility with the vehicle's electrical system and enables efficient power transfer to the load. The boost converter is controlled to maintain a stable DC bus voltage while allowing the fuel cell to operate within its optimal efficiency range. points can be moved from I_{sc} to V_{oc} . The output voltage of the fuel cell can be approximated as:

$$V_{fc} = E - V_{act} - V_{ohmic} - V_{conc}$$

where E is the reversible cell voltage, and V_{act} , V_{ohmic} , and V_{conc} represent activation, ohmic, and concentration losses, respectively.

The fuel cell output power is given by:

$$P_{fc} = V_{fc} \cdot I_{fc}$$

Since the fuel cell produces a variable and relatively low voltage, a unidirectional DC–DC boost converter is used to regulate and increase the voltage to match the DC bus level. The converter duty cycle D determines the output voltage as:

$$V_{out} = \frac{V_{fc}}{1 - D}$$

The fuel cell primarily supplies the base load, while maintaining operation within its optimal efficiency region. Similar modeling approaches are widely reported in hybrid vehicle studies [2], [6].

C. Integrated DC Bus and Power Contribution

Both the battery and fuel cell systems are connected to a common DC bus, which acts as the central node for power distribution within the vehicle. The DC bus supplies energy to various loads, including the electric motor and auxiliary systems. The integration of DC–DC converters ensures proper voltage regulation and efficient energy transfer from both sources.

This configuration allows the hybrid system to combine the advantages of both energy sources, where the fuel cell provides continuous power and the battery handles dynamic load variations. As a result, the overall system achieves improved efficiency, reliability, and adaptability under different operating conditions.

Both the battery and fuel cell are interfaced with a common DC bus through their respective DC–DC converters. The total power supplied to the load is the combined contribution of both sources:

$$P_{load} = P_{fc} + P_{bat}$$

This shared DC bus architecture ensures proper power distribution and voltage regulation across the vehicle system. The coordination between sources is governed by the energy management strategy, which determines the power split based on demand conditions. Such integrated configurations have been extensively explored in hybrid energy systems [1], [3], [10].

The developed MATLAB/Simulink model represents a hybrid electric vehicle energy system integrating a fuel cell, battery, DC–DC converters, and a common DC bus. The overall system is designed to ensure efficient power sharing between the energy sources based on load demand.

The fuel cell subsystem is modeled to provide continuous power and is connected to the DC bus through a unidirectional boost converter. This converter steps up the variable fuel cell voltage to a regulated DC bus voltage, ensuring stable operation. The battery subsystem is implemented using an equivalent circuit model and is interfaced with the DC bus via a bidirectional DC–DC converter. This allows the battery in both charging and discharging modes depending on system requirements.

A common DC bus acts as the central node where power

from both sources is combined and supplied to the vehicle load, which represents the traction motor and auxiliary components. The energy management system, implemented using a fuzzy logic controller, continuously monitors parameters such as load demand and battery state of charge. Based on these inputs, it generates control signals to regulate the converters and determine the power contribution from each source.

III. PROPOSED METHOD

A. Block diagram

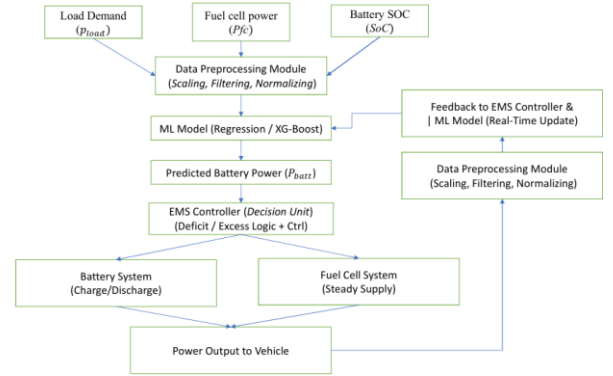


Fig. 5. Block diagram of Energy Management Strategy for Battery - Fuel Cell Hybrid Electric Vehicle with XGBoost-Based

The presented block diagram illustrates an intelligent energy management framework for a fuel cell–battery hybrid electric vehicle, integrating machine learning with real-time control. The system begins with three primary inputs: load demand (P_{load}), fuel cell power (P_{fc}), and battery state of charge (SOC). These inputs collectively represent the instantaneous operating condition of the vehicle and the availability of energy resources.

The input data is first processed through a data preprocessing module, where operations such as scaling, filtering, and normalization are applied. This step ensures that the data is conditioned appropriately for machine learning, improving numerical stability and prediction accuracy. Preprocessed data is then fed into a machine learning model, which may be implemented using regression techniques or advanced algorithms such as XGBoost. The ML model is trained to capture the nonlinear relationship between system inputs and the required battery power. Based on this learned behavior, it predicts the optimal battery power (P_{batt}) necessary to meet the load demand efficiently while maintaining system constraints. This predictive capability enhances the responsiveness of the energy management system under dynamic conditions.

The predicted battery power is passed to the Energy Management System (EMS) controller, which acts as the decision-making unit. The EMS evaluates the system state and classifies it into operating modes such as excess power or deficit power. Based on this classification, control signals are generated to regulate the operation of both the battery system and the fuel cell system.

The battery system operates in charge or discharge mode depending on the EMS decision. It supports peak load

conditions by supplying additional power or absorbs excess energy to maintain balance. Meanwhile, the fuel cell system provides a steady and continuous power supply, typically handling the base load demand to ensure efficient operation.

The combined output from both energy sources is delivered to the vehicle load through a common power output stage. Additionally, a feedback loop is incorporated, where real-time system data is continuously sent back to the preprocessing and machine learning modules. This enables adaptive learning and real-time updating of the model, improving system performance over time. Overall, the architecture demonstrates coordinated and intelligent control strategy that enhances energy efficiency, reliability, and adaptability in hybrid electric vehicle applications.

B. Flowchart

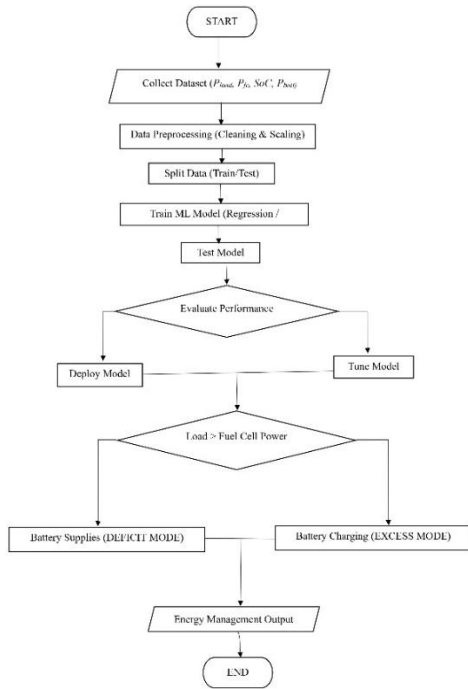


Fig. 6. Energy manage System Flow chart

The proposed system initiates with the acquisition of key operational parameters, including load demand (P_{load}), fuel cell power output (P_{fc}), and battery state of charge (SOC). These parameters represent the real-time operating condition of the hybrid electric vehicle and are essential for effective energy management. The collected data is passed through a preprocessing stage, where techniques such as normalization and scaling are applied to ensure uniformity and improve numerical stability. The processed dataset is then partitioned into training and testing subsets to facilitate the development and validation of the machine learning model.

Following preprocessing, a machine learning model, such as Extreme Gradient Boosting (XGBoost), is trained to learn the nonlinear relationship between input features and the required battery power (P_{bat}). The objective of the model is to approximate a function of the form:

$$P_{bat} = f(P_{load}, P_{fc}, SOC)$$

Once trained, the model is evaluated using performance

metrics such as mean squared error (MSE) or root mean square error (RMSE). After achieving satisfactory accuracy, the model is deployed for real-time prediction, enabling fast and reliable estimation of battery power requirements under dynamic operating conditions. The energy management system (EMS) utilizes the predicted battery power along with real-time measurements to determine the operating mode of the system. A power balance condition is established as:

$$P_{load} = P_{fc} + P_{bat}$$

Based on this relationship, the EMS identifies whether the system is in a deficit or excess power condition. In deficit mode, where the load demand exceeds the fuel cell output:

$$P_{load} > P_{fc} \Rightarrow P_{bat} = P_{load} - P_{fc}$$

the battery is commanded to discharge and supply the additional required power. Conversely, in excess mode, where the fuel cell generates more power than required by the load:

$$P_{fc} > P_{load} \Rightarrow P_{bat} = -(P_{fc} - P_{load})$$

The surplus energy is directed towards charging the battery.

This bidirectional energy flow ensures optimal utilization of both energy sources. By integrating predictive modeling with real-time control, the proposed system enhances efficiency, reduces stress on individual components, and enables adaptive power distribution in hybrid electric vehicle applications.

C. Ratings of the components to develop the ML model for the Energy management system

Component	Parameter	Rating / Value
Battery	Energy	99.8wAh
Battery	Nominal voltage	552 v
Battery	Capacity	181Ah
Fuel cell	Peak Power	70kW
Fuel cell	Output voltage	250-350v

TABLE 1: Electrical Specifications and Component Ratings of the Proposed Hybrid Energy Management System

The table-1 presents the key electrical specifications and component ratings considered for the design and simulation of the proposed fuel cell–battery hybrid energy management system. The battery system is characterized by a high energy capacity of 99.8 kWh, a nominal voltage of 552 V, and a capacity of 181 Ah, which enables it to support peak power demands and provide energy storage for dynamic operating conditions. In parallel, the fuel cell system is designed to deliver a peak power output of 70 kW, ensuring sufficient base load supply for the vehicle.

The fuel cell operates within an output voltage range of 250–350 V, which is further regulated and stepped up using a DC–DC boost converter to match the DC bus requirements.

This combination of high-energy battery storage and moderate fuel cell power rating ensures an optimal balance between energy density and power density, allowing efficient load sharing under varying demand conditions. The selected ratings are aligned with typical hybrid electric vehicle architectures, where the fuel cell provides steady-state power while the battery compensates for transient load variations and energy buffering. Such a coordinated design enhances system efficiency, reduces component stress, and improves overall performance of the hybrid energy system [11].

IV. RESULT AND DISCUSSION

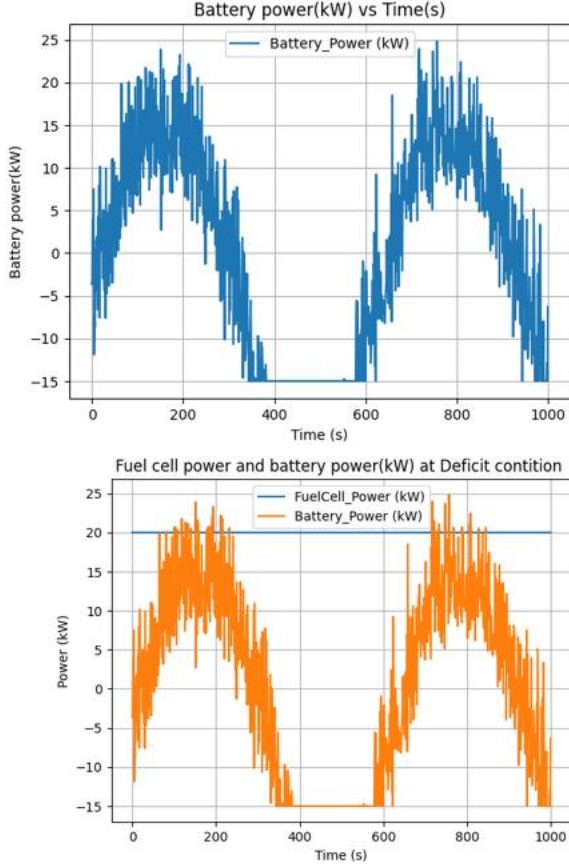


Fig. 7. Power Distribution Characteristics of Fuel Cell–Battery Hybrid System under Deficit Operating Condition

Fig. 7 presents the performance of the hybrid fuel cell–battery system under deficit operating conditions. In this mode, the load demand exceeds the power generated by the fuel cell, resulting in a power imbalance. The fuel cell operates at a relatively constant output, supplying the base load, while the battery compensates for the deficit by discharging. This is indicated by the positive battery power values observed in the plot. The combined power from both sources satisfies the load demand; however, significant fluctuations are evident in the battery power profile. These oscillations indicate that the system responds reactively to load variations, which may lead to reduced efficiency and increased stress on the battery due to frequent charge–discharge cycles [12], [13].

However, the battery power waveform exhibits noticeable fluctuations, reflecting the system’s dependence on reactive control mechanisms. These oscillations occur due to rapid variations in load demand and the absence of

predictive control, causing frequent adjustments in battery output. Such behavior may lead to increased stress on the battery, reduced efficiency, and potential degradation over time. The results highlight the limitations of conventional energy management strategies and emphasize the need for an intelligent predictive approach to achieve smoother power distribution and improved system performance.

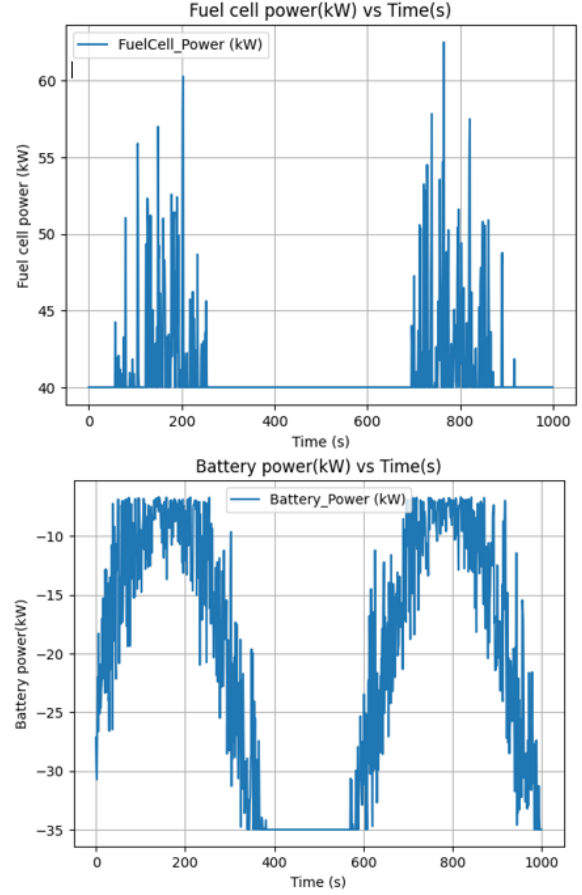


Fig. 8. Power Distribution Characteristics of Fuel Cell–Battery Hybrid System under Excess Operating Condition

Fig. 8 illustrates the performance of the hybrid fuel cell–battery system under excess operating conditions, where the power generated by the fuel cell exceeds the instantaneous load demand. In this operating mode, the fuel cell continues to function near its nominal operating point, delivering a steady output to maintain high efficiency. The surplus energy generated is effectively redirected to the battery for storage, which is indicated by the negative battery power values representing the charging mode. This energy flow strategy ensures that excess power is not dissipated and is instead utilized for future load requirements, thereby improving overall system energy utilization [14], [15].

The combined power profile confirms that the total power supplied by the fuel cell and battery remains balanced with the load demand. However, noticeable fluctuations are present in both fuel cell output and battery charging power due to the absence of predictive control mechanisms. These variations arise from the system’s reliance on instantaneous feedback, leading to dynamic. Such non-uniform energy distribution can reduce charging efficiency and impose additional stress on the battery system over time. These limitations highlight the need for advanced energy management approaches, such as machine learning-based predictive control, which can minimize power oscillations and enhance system stability and performance [16], [17].

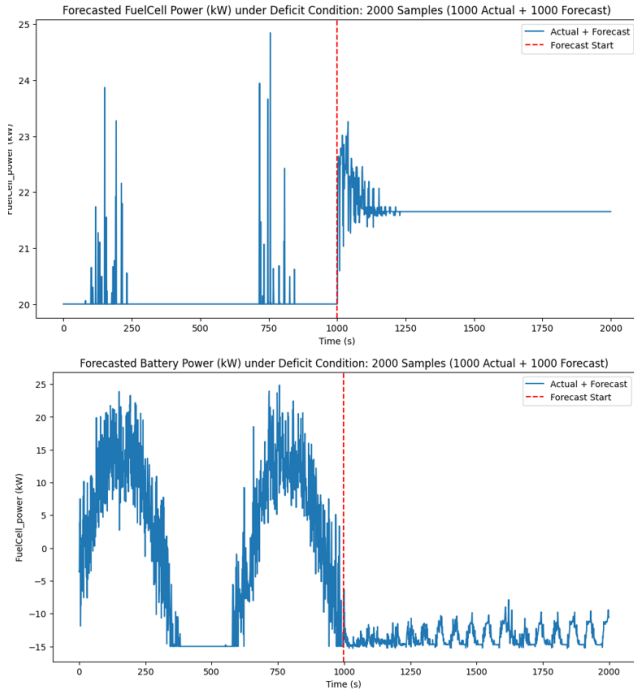


Fig. 9. Predicted Power Profiles of Fuel Cell and Battery under Excess Operating Condition Using XGBoost Algorithm

Fig. 9 presents the predicted performance of the hybrid fuel cell–battery system under deficit operating conditions using the XGBoost algorithm. The figure shows both actual and forecasted values of fuel cell power and battery power over time. After the forecast initiation point, indicated by the dashed line, the predicted fuel cell power stabilizes around a consistent operating level, demonstrating the model’s ability to maintain a steady base power output. This behavior indicates that the machine learning model effectively captures the underlying trend of the fuel cell operation and reduces unnecessary fluctuations compared to conventional control methods. Such stable operation is essential for improving fuel cell efficiency and ensuring reliable long-term performance [18], [19].

Similarly, the battery power prediction exhibits a smoother and more controlled profile after the forecasting point. The model successfully anticipates the required battery contribution during deficit conditions, minimizing abrupt variations and reducing oscillatory behavior observed in earlier results. This indicates that the XGBoost model enhances the dynamic response of the system by providing accurate power estimates in advance. As a result, the battery operates more efficiently with reduced stress from frequent charge–discharge transitions. Overall, the predictive approach improves power distribution, enhances system stability, and demonstrates the effectiveness of machine learning-based energy management in hybrid systems [20], [21].

Fig. 11 illustrates the predicted performance of the hybrid fuel cell–battery system under excess operating conditions using the XGBoost algorithm. The figure presents both actual and forecasted power profiles of the fuel cell and battery over time. After the forecast initiation point, indicated by the dashed line, the fuel cell power stabilizes at a nearly constant level, reflecting the model’s capability to maintain steady-state operation. This indicates that the XGBoost model effectively learns the operational characteristics of the fuel cell and minimizes unnecessary fluctuations.

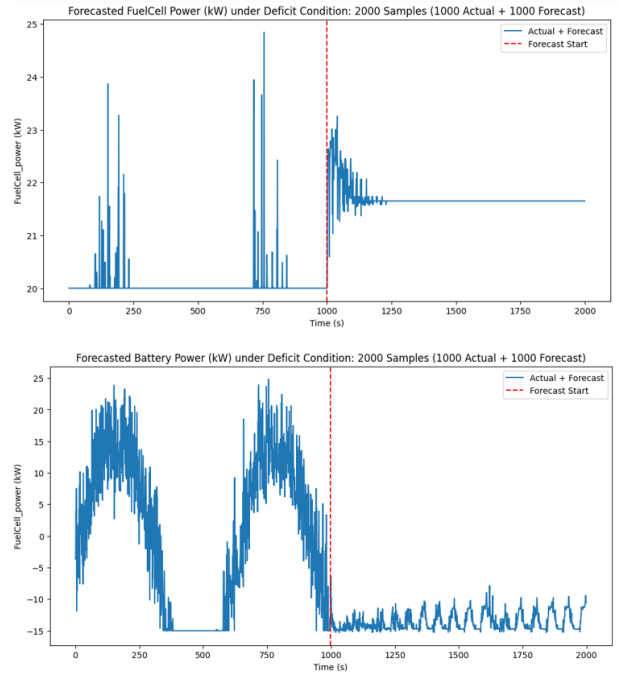


Fig. 9. XGBoost-Based Predicted Power Profiles of Fuel Cell and Battery under Excess Operating Condition

Such stabilization is beneficial for maintaining optimal efficiency and reducing dynamic stress on the fuel cell system [22], [23]. In addition, the battery power profile shows a smooth transition into a controlled charging mode under excess conditions. The predicted battery power exhibits reduced oscillations compared to earlier results, indicating improved stability in energy storage behavior. The negative battery power values confirm that surplus energy from the fuel cell is being efficiently stored in the battery. This predictive control approach enables better energy utilization, reduces system losses, and enhances overall performance. The results demonstrate that the integration of machine learning significantly improves power distribution and system reliability in hybrid energy management systems [24], [25].

CONCLUSION

This work presented an intelligent energy management strategy for a fuel cell–battery hybrid electric vehicle by integrating fuzzy logic control with a machine learning-based prediction model. The hybrid system was modeled and implemented in MATLAB/Simulink, where the fuel cell and battery were coordinated through DC–DC converters and a common DC bus. Initially, a conventional control approach was analyzed under both deficit and excess operating conditions, which revealed noticeable fluctuations in power distribution and reactive behavior of the system. These limitations highlighted the need for a more adaptive and predictive control strategy.

To address these challenges, an XGBoost-based machine learning model was developed to predict battery power requirements using key system parameters such as load demand, fuel cell power, and battery state of charge. The results demonstrated that the proposed predictive approach significantly improved system performance by reducing power oscillations and enabling smoother energy distribution. Under both deficit and excess conditions, the fuel cell maintained stable operation while the battery exhibited controlled charging and discharging behavior. This

not only enhanced overall efficiency but also reduced stress on system components, thereby improving reliability and operational lifespan.

Overall, the integration of machine learning with conventional control methods proved to be an effective solution for advanced energy management in hybrid electric vehicles. The proposed system successfully achieved balanced power sharing, improved adaptability to varying load conditions, and efficient utilization of energy sources. Future work can focus on real-time hardware implementation, incorporation of additional parameters such as temperature and driving patterns, and the application of advanced deep learning techniques to further enhance prediction accuracy and system performance.

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